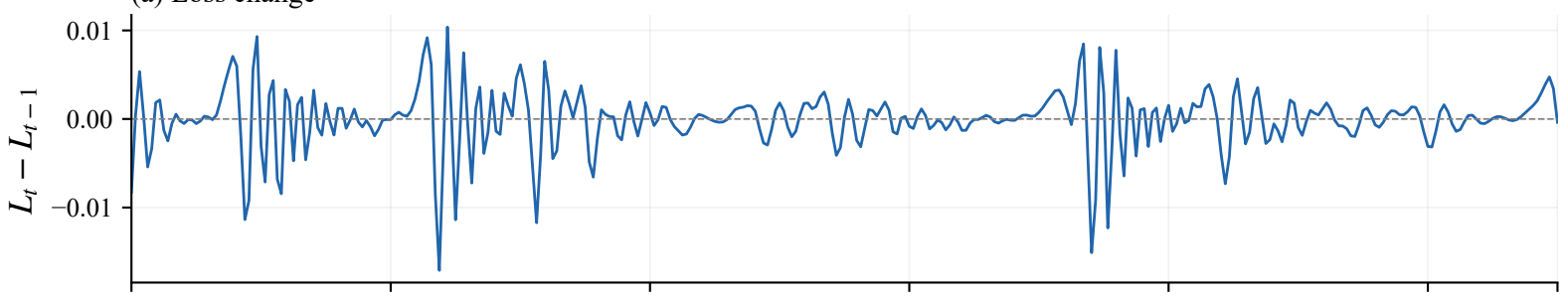
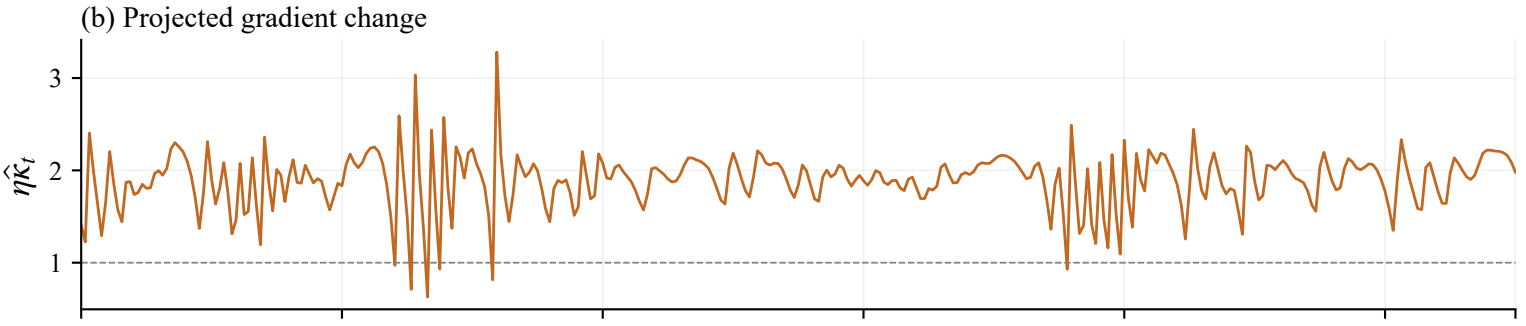
CIFAR-10 MLP: unchanged Adam control

Full 5,000-example objective; every state from 448 to 800

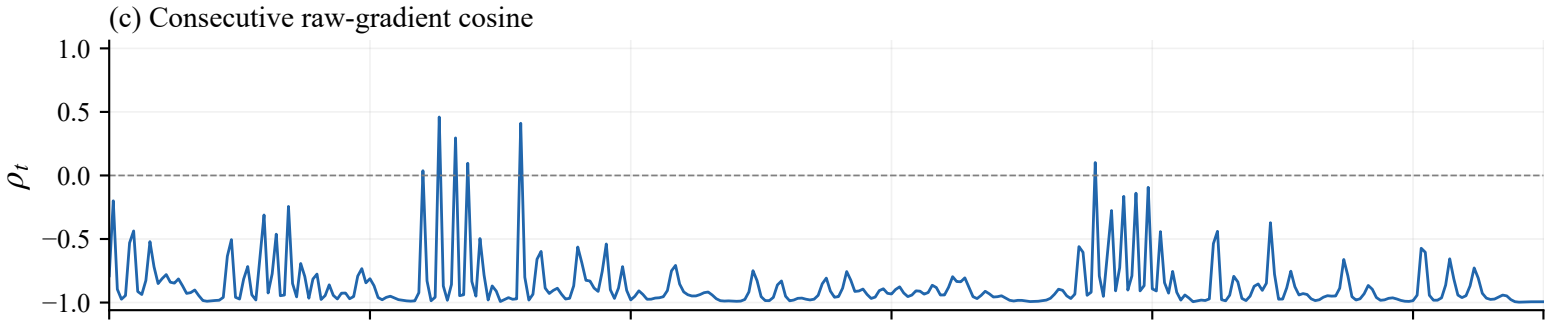
(a) Loss change



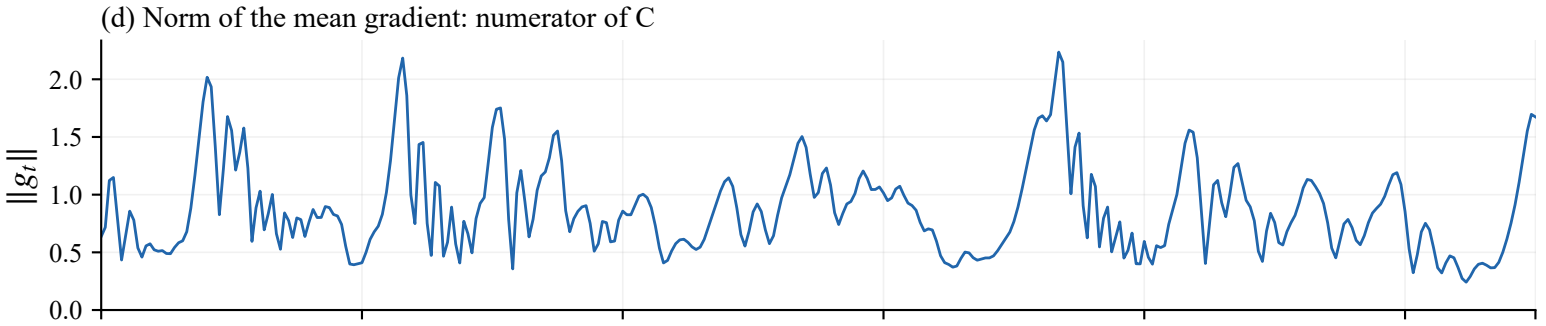
(b) Projected gradient change



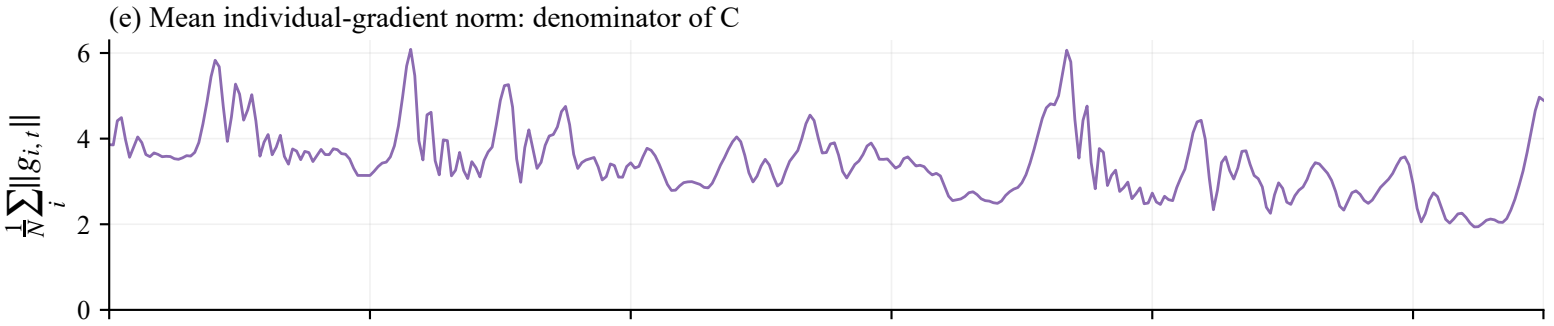
(c) Consecutive raw-gradient cosine



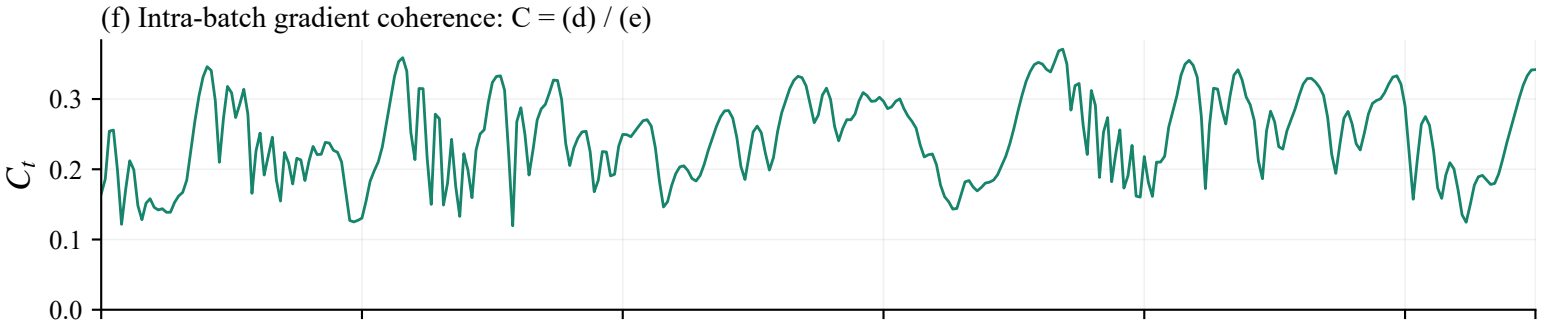
(d) Norm of the mean gradient: numerator of C



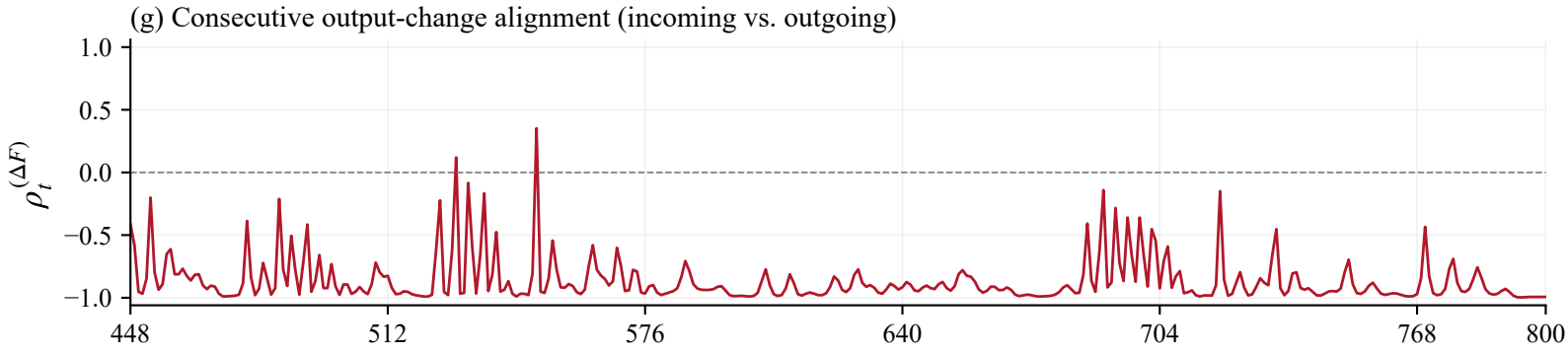
(e) Mean individual-gradient norm: denominator of C



(f) Intra-batch gradient coherence: C = (d) / (e)



(g) Consecutive output-change alignment (incoming vs. outgoing)



$$g_t = \frac{1}{N}\sum_i g_{i,t}, \qquad C_t = \frac{\|g_t\|}{\frac{1}{N}\sum_i \|g_{i,t}\|} = \frac{\text{panel (d)}}{\text{panel (e)}}$$

$$\Delta F_t = F_{t+1} - F_t, \qquad \rho_t^{(\Delta F)} = \cos(\Delta F_t, \Delta F_{t-1})$$

Raw gradients include all weights and biases. No smoothing or subsampling.